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The rapid permeation of AI into nearly all walks of life and every profession calls for innovative approaches of public AI literacy and education in all settings. This project will design, develop, and test a sequence of lessons for high school aged youth from east Tennessee that will teach them artificial intelligence concepts by using images from planetary exploration and emerging Generative AI tools to create visually appealing generative artworks and digital stories. AI concepts grounded in topics that are relevant and engaging through the lens of creative works can serve as an invitation for underrepresented and under-resourced youths to explore fields that they would not otherwise choose or have the opportunity to learn about. The project team including AI researchers and educators, STEM disciplinary experts and the broader creative works community will employ a mixture of out-of-school and online activities, integrating Planetary Science, planetary exploration, big data, creative works, and Generative AI. The works created by youths with the assistance of Generative AI and validated by scientists on planetary exploration and Planetary Science will be showcased online, at the American Museum of Science and Energy, and other public places. This Integrating Research & Practice project is funded by the Advancing Informal STEM Learning (AISL) program, which supports projects that: (a) contribute to research and practice that considers informal STEM learning's role in equity and belonging in STEM; (b) promote personal and educational success in STEM; (c) advance public engagement in scientific discovery; (d) foster interest in STEM careers; (e) create and enhance the theoretical and empirical foundations for effective informal STEM learning; (f) improve community vibrancy; and/or (g) enhance science communication and the public's engagement in and understanding of STEM and STEM processes, and co-funded by the Innovative Technology Experiences for Students and Teachers (ITEST) program, which supports projects that build understandings of practices, program elements, contexts and processes contributing to increasing students' knowledge and interest in science, technology, engineering, and mathematics (STEM) and information and communication technology (ICT) careers.

Over the three-year project duration, the project will directly impact approximately 360 high school youth and will reach out to a much larger number of participants through online webinars, workshops, newsletters, and exhibits. The project will investigate four research questions: (1) How does participation in an interdisciplinary out-of-school STEM program influence high school youths' AI knowledge and skills, STEM engagement and identity, and sense of belonging to a scientific community? (2) What do youths learn about the value of their personal and cultural knowledge and skills (i.e., assets) to the local and broader community? (3) What program challenges, features, and content appear to have the greatest influence on engagement, identity, and belonging? (4) How do outcomes vary by youth demographics including gender, ethnicity, and urban/rural environment? The research will use a mixed-methods design, combining quantitative surveys with qualitative written reflections and pictures with commentary. In the final year of the project, professional development workshops will train educators to use learning materials generated from the project. Research findings produced through this work will contribute new knowledge on methods and learning designs in integrating AI, STEM and art for informal STEM education.